Product Alpha R7 Architecture Overview

Aster Loop (synthetic organization) | Product Alpha R7 | 2026-07-13 | p95 184 ms | ADR-042

System flow figure

Clients	→	Edge API	→	Release Coordinator	→	Event Bus
Release ticket		Validate intent		Durable admission		Versioned event

Decision boundary

ADR-042 returns only after coordinator durable admission. The coordinator writes the decision store; Event Bus and Observability consume versioned evidence without changing a release decision.

Supporting path

Coordinator → Decision Store; Edge API → Observability; Event Bus → Observability. Clients own intent and idempotency, while Release Engineering owns transition legality.

Latency evidence

The end-to-end Product Alpha R7 p95 budget is 184 ms: edge 28, API validation 32, durable admission 46, event publish 24, observability 18, and reserve 36.

Measurement method

Measure from edge acceptance through durable ticket response. The synthetic R7 cohort excludes cold starts and keeps queue depth below 200. Warn at 165 ms for five minutes; page at 184 ms for ten minutes.

Contract and rollout controls

Versioned requested.v1 and decided.v1 events hold synthetic release metadata. The request idempotency key returns the original ticket and never republishes a decision. Timeout is a terminal state, never approval.

Release sequence

Shadow evaluate, then enable 10%, 50%, and 100%. Stop for sustained p95 above 184 ms, admission error above 0.2%, or a schema guard failure. Routing rollback does not rewrite durable decisions.